

Original Article

Performance Evaluation of Fly Ash-Based Geopolymer Concrete

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Abstract – The increasing environmental impact of Portland cement production has encouraged the development of sustainable construction materials. This study investigates the performance of fly ash-based geopolymer concrete as an alternative binder system. Fly ash, an industrial by-product, was activated using alkaline solutions comprising sodium hydroxide and sodium silicate. The effects of parameters such as molarity of alkaline solution, water-to-binder ratio, and curing conditions were analyzed. Experimental findings indicate that geopolymer concrete exhibits superior compressive strength, enhanced durability, and reduced permeability compared to conventional concrete. Heat curing significantly improves early strength development. The study concludes that fly ash-based geopolymer concrete is a viable and eco-friendly alternative for modern construction

Keywords – *Geopolymer concrete, Fly ash, Alkali activation, Compressive strength, Sustainable materials*

1. INTRODUCTION:

Concrete is the most widely used construction material, primarily composed of Portland cement as a binder. However, cement production contributes significantly to global carbon dioxide emissions, posing serious environmental challenges. The demand for sustainable alternatives has led researchers to explore industrial by-products such as fly ash.

Fly ash, generated from coal-fired power plants, is abundantly available and often underutilized. Its pozzolanic nature makes it suitable for use in concrete. Geopolymer technology utilizes fly ash as a primary binder through alkali activation, eliminating the need for Portland cement. This process forms a three-dimensional alumino-silicate network that provides high strength and durability.

This study focuses on evaluating the performance characteristics of fly ash-based geopolymer concrete under different mix and curing conditions.

2. LITERATURE REVIEW:

Previous research has highlighted the potential of geopolymer concrete as a sustainable construction

material. Studies have shown that compressive strength increases with higher concentrations of alkaline activators. The ratio of sodium silicate to sodium hydroxide significantly influences mechanical properties.

Elevated curing temperatures accelerate geopolymerization, resulting in higher early strength. Additionally, geopolymer concrete demonstrates excellent resistance to sulfate attack, reduced shrinkage, and improved durability compared to conventional concrete. However, variations in material properties and lack of standardized design procedures require further investigation.

3. MATERIALS AND MATHODS:

A. Materials:

The materials used in this study include:

- Low-calcium (Class F) fly ash
- Fine aggregate (river sand)
- Coarse aggregate (crushed basalt)
- Sodium hydroxide (NaOH)
- Sodium silicate (Na_2SiO_3)
- Water

B. Mix Proportions:

Concrete was prepared using a nominal mix ratio of 1:1.5:3. The alkaline solution-to-fly ash ratio and water-to-binder ratio were varied to analyze their effects.

C. Preparation of Alkaline Solution:

Sodium hydroxide solutions of different molarities were prepared by dissolving NaOH flakes in water. Sodium silicate solution was then mixed with NaOH to form the alkaline activator.

D. Mixing and Casting:

Dry materials were mixed thoroughly before adding the alkaline solution. The mixture was poured into molds and compacted properly.

E. Curing Methods:

Two curing methods were adopted:

1. Ambient curing
2. Oven curing at elevated temperatures (up to 90°C)

F. Testing Procedures

The following tests were conducted:

- Compressive strength test
- Flexural strength test
- Split tensile strength test
- Workability test (slump test)

4. RESULTS AND DISCUSSION:

A. Effect of Alkaline Concentration: An increase in sodium hydroxide molarity led to higher compressive strength due to enhanced geopolymerization.

B. Effect of Water-to-Binder Ratio: Higher water content resulted in reduced compressive strength consistent with conventional concrete behavior.

C. Effect of Curing Temperature: Oven-cured specimens exhibited significantly higher strength compared to ambient-cured samples due to accelerated chemical reactions.

D. Durability Performance: Geopolymer concrete demonstrated:

- Reduced permeability
- Improved resistance to chemical attack
- Lower heat of hydration
- Enhanced long-term durability

5. ADVANTAGES OF GEOPOLYMER CONCRETE:

- Reduced CO₂ emissions
- Utilization of industrial waste (fly ash)
- Improved workability
- High early strength
- Better resistance to sulfate and acid attack
- Reduced thermal cracking

6. LIMITATIONS:

- Requirement of controlled curing conditions
- Handling of alkaline solutions requires safety precautions
- Limited availability of design standards

7. CONCLUSION:

Fly ash-based geopolymer concrete is a sustainable and high-performance alternative to conventional concrete. The elimination of Portland cement significantly reduces environmental impact. Experimental results confirm that proper control of mix proportions and curing conditions leads to improved mechanical and durability properties. The adoption of geopolymer technology can contribute to sustainable infrastructure development and reduced carbon emissions in the construction industry.

REFERENCES:

- [1] [J. Davidovits, Geopolymer Chemistry and Applications, 2002.
- [2] D. Hardjito and B. V. Rangan, "Development and properties of low-calcium fly ash-based geopolymer concrete," 2005.
- [3] S. Wallah and B. V. Rangan, "Low-calcium fly ash-based geopolymer concrete," 2006.
- [4] A. Palomo and A. Jiménez, "Alkali-activated fly ash binders," 2011.
- [5] N. Bhosale and D. Shinde, "Studies on geopolymer concrete," 2012.